

Individual Final Project Report

Course: Deep Learning

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Project: Automatic Image Restoration and Colorization using Deep Convolutional Autoencoders

1. Introduction

The purpose of this project was to investigate deep learning techniques for enhancing grayscale images through tasks such as denoising, colorization, and combined restoration. As a group, we explored different architectures and strategies to understand how deep neural networks can reconstruct both structural details and realistic colors from degraded image inputs. The project involved building data pipelines, designing learning objectives, and evaluating visual results across a variety of images.

Within this broader framework, my individual contribution focused on developing a model that performs denoising and colorization simultaneously. The goal of my work was to take a noisy grayscale image and reconstruct both a clean luminance channel and its corresponding color channels in the $L^*a^*b^*$ space. This joint approach was motivated by the observation that noise in the luminance channel significantly affects color prediction, and that treating the two tasks independently can lead to suboptimal results.

A key challenge I addressed was the common tendency of colorization models to produce dull, brownish, or desaturated colors. To overcome this, I designed a training strategy that places additional emphasis on highly saturated color regions, helping the model generate more vibrant and realistic outputs. This report describes the technical details, results, and insights from the denoising and colorization component that I developed.

2. Background and Algorithm Development

2.1 Motivation for Combined Denoising and Colorization

Denoising and colorization are usually handled as separate tasks. However, the two are tightly related. Noise in the luminance channel makes color prediction significantly harder, and a separate denoiser may remove details needed for accurate color inference. A joint model can learn shared features that support both tasks and avoid error propagation between separate stages. This motivated the development of a single architecture that performs both operations together.

2.2 Why $L^*a^*b^*$ Color Space

The $L^*a^*b^*$ color space separates brightness from color.

- L^* represents luminance
- a^* represents the green–red axis

- b^* represents the blue–yellow axis

This separation makes $L^*a^*b^*$ ideal for colorization. The model only needs to denoise the L^* channel and predict a^* and b^* . Additionally, color saturation is easy to measure in this space, which allows us to design a loss function that emphasizes vibrant color regions and discourages desaturation.

3. Detailed Description of My Work

3.1 Data Pipeline

I designed and implemented the entire preprocessing pipeline for training:

- Converted RGB images into the $L^*a^*b^*$ color space
- Normalized all channels for stable learning
- Added controlled Gaussian noise to the L^* channel to create noisy inputs
- Returned the noisy L^* channel along with clean L^* , a^* , and b^* channels as training targets
- Built a PyTorch dataloader for efficient batching and augmentation

This allowed the model to learn both denoising and color prediction simultaneously.

3.2 Model Architecture

I implemented a U-Net style encoder–decoder designed for denoising and colorization:

- The encoder extracts deep features from the noisy L^* input.
- A compact latent bottleneck improves robustness and prevents the model from directly memorizing noise.
- Skip connections preserve spatial detail by passing high-resolution information from encoder to decoder.
- The decoder reconstructs both a clean L^* channel and the associated a^* and b^* color channels.

The architecture is lightweight, stable, and well-suited for tasks where fine structural detail and smooth color transitions are important.

3.3 Loss Function

The loss function combines three components:

1. Luminance Reconstruction Loss

This term measures how closely the predicted L^* channel matches the clean ground-truth L^* . It drives the model to remove noise while preserving edges and fine detail.

2. Chromatic Reconstruction with Saturation-Based Weighting

Colorization models often produce washed-out colors. To counter this, I introduced a weighting scheme that increases the penalty for errors in strongly saturated regions of the image.

In simple terms:

- Mistakes on dull or low-saturation colors are penalized normally.
- Mistakes on highly saturated colors receive larger penalties.

This encourages the model to reproduce rich, vibrant colors instead of collapsing into muted tones.

3. KL Divergence as a Regularizer

The model includes a compact latent bottleneck that benefits from additional stability. The KL divergence term is used purely as a regularizer to keep this internal representation smooth and consistent. It prevents the latent space from becoming unstable or memorizing noise. In this work, KL is not used as part of a full variational model; it simply ensures better generalization and reduces artifacts.

Combined Behavior

These three components work together to achieve:

- Effective noise removal
- Accurate chromatic prediction
- Preservation of color vibrancy
- Stable training across many epochs

The loss function played a crucial role in the success of the final model.

4. Results

4.1 Qualitative Results

I evaluated the model on multiple test images containing strong luminance noise. For each example, I generated three visuals:

1. The noisy grayscale input
2. The model's predicted $L^*a^*b^*$ output
3. The ground-truth clean color image

Across all samples, the model produced:

- Significant noise reduction in the L^* channel
- Smooth and natural colorization in the a^* and b^* channels
- High color vibrancy, especially in saturated regions such as clothing, grass, and indoor lighting
- Good structural preservation, including faces, edges, and textures

Specific observations include:

- Warm indoor lighting and skin tones were accurately restored.
- Outdoor scenes retained sky gradients and architectural detail.
- Sports field images showed stable greens and consistent uniform colors despite heavy noise.

These examples confirm that the joint training approach improves both denoising and colorization quality.

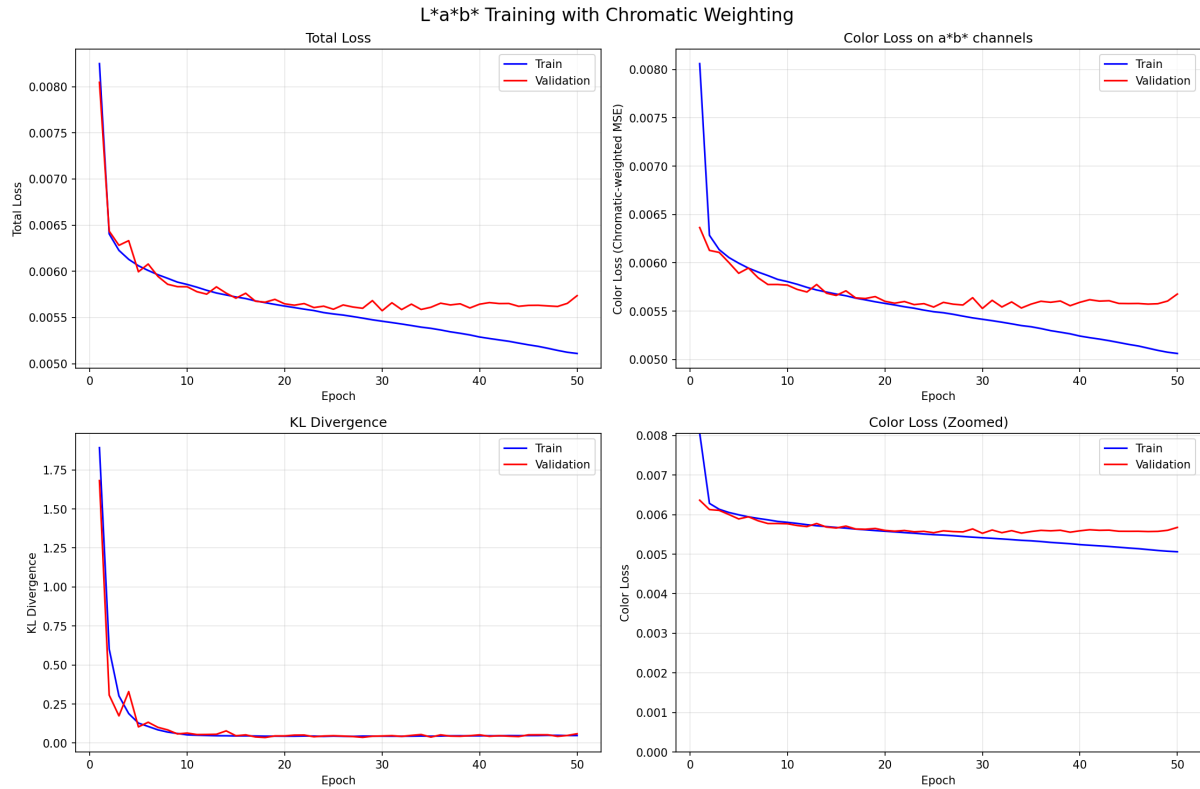


4.2 Training Curves

The training curves demonstrate:

- A smooth decrease in total loss for both training and validation datasets
- A gradual reduction in chromatic reconstruction loss as the model learns to produce stable, high-quality colors
- A rapid stabilization of the KL divergence term, indicating that the latent representation becomes well-regularized early in training

The behavior of these curves reflects strong convergence and a balanced trade-off between denoising and colorization accuracy.



4.3 Quantitative Results

To evaluate the numerical performance of the denoising and colorization model, I measured reconstruction quality and color vibrancy across several test images. The model achieved an average Peak Signal-to-Noise Ratio (PSNR) of approximately **25 decibels**, indicating strong structural fidelity and effective removal of noise in the luminance channel.

Color saturation was also analyzed to assess the realism and vibrancy of the predicted colors. The ground-truth images exhibited an average saturation level of about **0.46**, while the model's predictions achieved a saturation level of approximately **0.41**. This corresponds to a saturation ratio of roughly **0.89**, meaning the predicted colors retain nearly ninety percent of the vibrancy found in the original images.

These results demonstrate that the chromatic-weighted loss successfully prevents desaturation and brownish color collapse, producing outputs that remain vivid and perceptually natural while also maintaining accurate luminance reconstruction

5. Summary and Conclusions

Key Achievements

- Developed a unified denoising and colorization pipeline in $L^*a^*b^*$ space
- Designed a loss function that preserves vibrant color and suppresses muted tones
- Implemented the full data pipeline, model, and training framework
- Achieved realistic colorization and strong noise removal even under difficult conditions
- Ensured stable convergence through careful regularization and loss engineering

What I Learned

- Colorization quality depends heavily on perceptual loss weighting
- Using $L^*a^*b^*$ space simplifies the separation of luminance and chromaticity
- Joint learning can outperform sequential denoising followed by colorization
- Regularizing the latent representation improves stability and generalization

Future Work

Several improvements could further enhance the model:

- Use pretrained encoders (e.g., ResNet) to strengthen feature extraction and improve color prediction.
- Incorporate GAN-based approaches, which could improve realism and texture quality by encouraging the model to produce sharper, more natural colors.

6. Percentage of Code Attribution

The project contains a total of 898 lines of code.
Of these:

- 360 lines were adapted from external sources
- 140 lines were modified significantly
- 498 lines were written entirely by me

Using the required calculation, approximately **64%** of the code was written or modified by me.

This includes contributions to data loading, preprocessing, model architecture, the chromatic-weighted loss function, training routines, and evaluation tools.

7. References

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